# ERIC HOU

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## EDUCATION

*PURDUE UNIVERSITY, West Lafayette, IN* — *Present - Jan 2029 (Expected)*

- Bachelor of Science | College of Engineering

*MISSION SAN JOSE HIGH SCHOOL, Fremont, CA* — *Aug 2021 - Jun 2025*

- GPA: 4.38 Weighted | Advanced Coursework: Multivariable Calculus, Linear Algebra, Statistics, Physics Mechanics + E&M
- SAT Score: 1550

## RESEARCH

**First Author - IEEE Paper** — *Sep 2023 - Dec 2024*

- Engineered a vertical wall-scaling quadrupedal robot with adaptive suction for infrastructure monitoring with 9-lb payload
- Published and presented findings as first author at the 2024 IEEE IRC Conference

**Aspiring Scholars Directed Research Program - Student Researcher** — *Aug 2021 - Jun 2023*

- Selected for prestigious research program with 15% acceptance rate in that term's cohort
- Applied Quantum Annealing techniques to isolate particle collision jets using regression models and neural nets w/ Dr. McMahan

## PROJECTS

**5-Stage Electromagnetic Coilgun** — *Jan 2025 - May 2025*

- Engineered multi-stage coilgun using Fusion 360 and KiCAD for high-current supercapacitor discharge control PCB
- Optimized magnetic flux using MATLAB simulations and Fusion 360
- Implemented rotating base and synchronized triggering for optimized projectile velocity, achieving 1.7m range with a 1lb payload

**Autodesk Fusion 360 Wire Generator Plugin** — *Oct 2023 - Feb 2024*

- Developed python script using Fusion 360 API to generate parameterized spline paths and wiring layouts
- Published plugin on Autodesk Store → 3,000+ active users, 5-star rating

## EXPERIENCES

**Avionics Engineer** — *Aug 2025 - Present*

*Active Controls | Purdue Space Program (PSP)*

- Designed power monitor & control board for ASTRA, a self-stabilizing and landing EDF aerial vehicle

**Hardware/Electrical Engineer** — *Aug 2025 - Present*

*Purdue Embedded Systems | SlayterHIL Project*

- Designing hardware-in-the-Loop (HIL) testing infrastructure for drone flight controllers, simulating sensors and motor signals
- Prototyping communications and power distribution PCBs to interface between DUT and test node hardware

**FTC TEAM 13356** — *Jul 2021 - Jun 2025*

*Hardware Co-captain*

- Directed end-to-end hardware design, manufacturing, and testing for competitive robotics seasons
- **Achievements:** #3 in California, Top 30 worldwide; qualified for Maryland Tech Invitational (top 0.5%) three years running
- Prototyped, manufactured, and shipped 300+ custom STEM kits for global organizations
- Developed and taught CAD/programming curriculum to Afghan refugees, Taiwanese Schools, and rookie teams
- Organized logistics for 8+ community events, reaching over 2000+ people

**LEARNIFY+ Nonprofit** — *Sep 2021 - Jan 2024*

*Head of Computer Science & Board of Director*

- Led a team of 50+ tutors delivering classes to 200+ students across 10 states and 4 countries
- Created course curricula in Java, Python, and HTML/CSS
- Volunteered 650+ hours in curriculum design, organizational leadership, and marketing

## OTHER

**SASA (Student Association for Stem Advocacy)** — *Feb 2023 - May 2025*

*Logistics Manager / Lobbyist*

- Coordinated advocacy campaigns and conferences; lobbied senators and congressional offices for STEM education funding

**HM Award - Toshiba ExploraVision Competition**

- Developed design and business proposal for eco-friendly air purifying system using algae photobioreactor modules

**MSJ Robotics Club President**

- Planned and hosted club meetings; coordinated outreach events and fundraisers; interviewed candidates

## SKILLS

**Design / Manufacturing:** KiCAD, FDM-3D Printing, Waterjet, CNC, Fusion 360 CAD/CAM/FEA

**Programming Languages:** Python, Java, C, HTML/JS/CSS